

# FPGA-Oriented Object Detection using Background Subtraction

RTL Design, Simulation and Verification

Tools Used:

Verilog HDL  
MATLAB  
Simulink  
Xilinx ISE

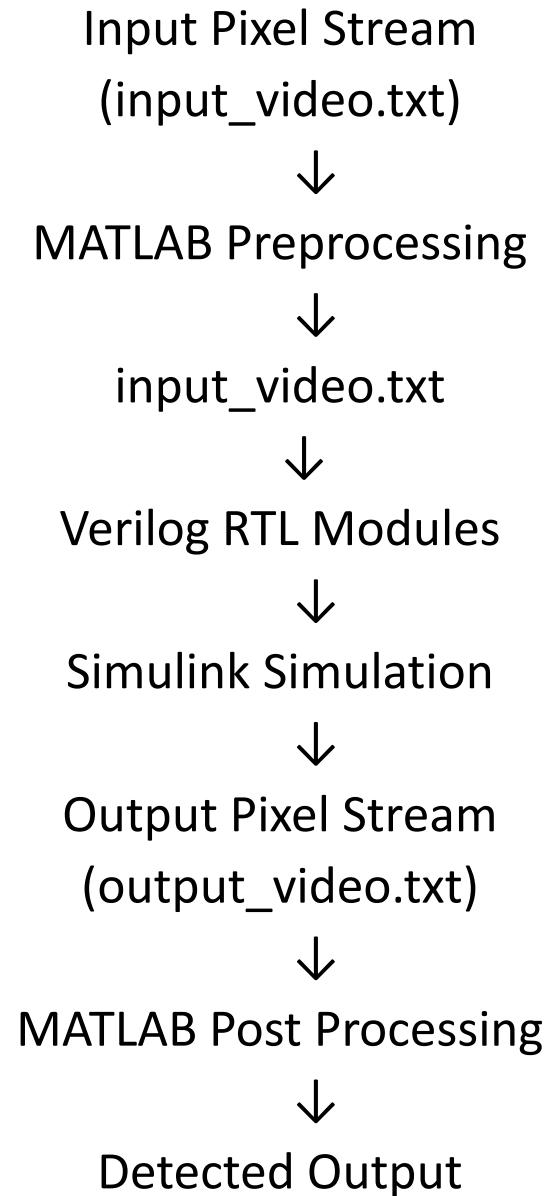
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# Problem Statement

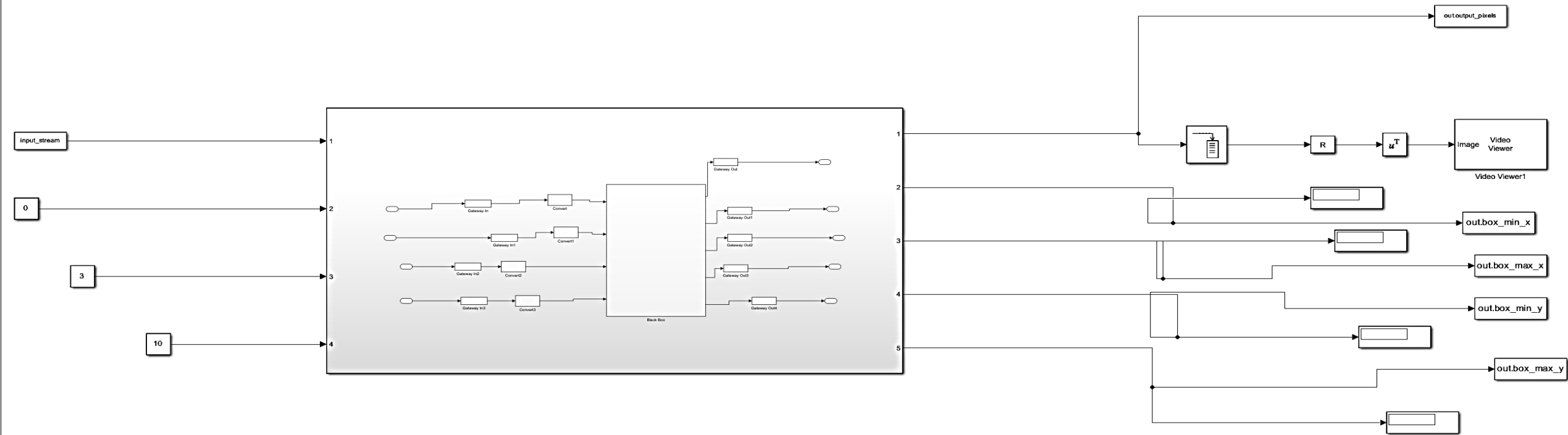
- Design and verify an RTL-based object detection pipeline capable of detecting moving objects using background subtraction.
- The objective was to understand how computer vision algorithms can be translated into RTL hardware modules and verified through simulation.



# Workflow

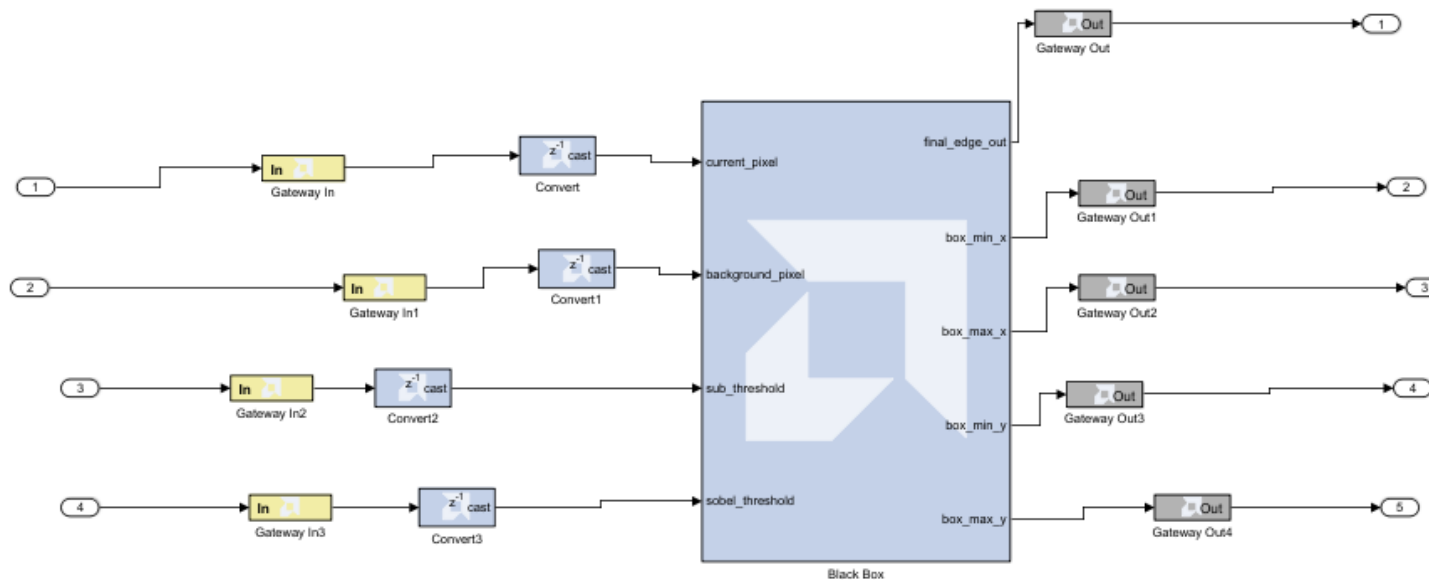


# RTL Simulation Model



The Verilog RTL modules were integrated into Simulink for functional verification and output visualization.

# RTL Processing Pipeline



## RTL Modules

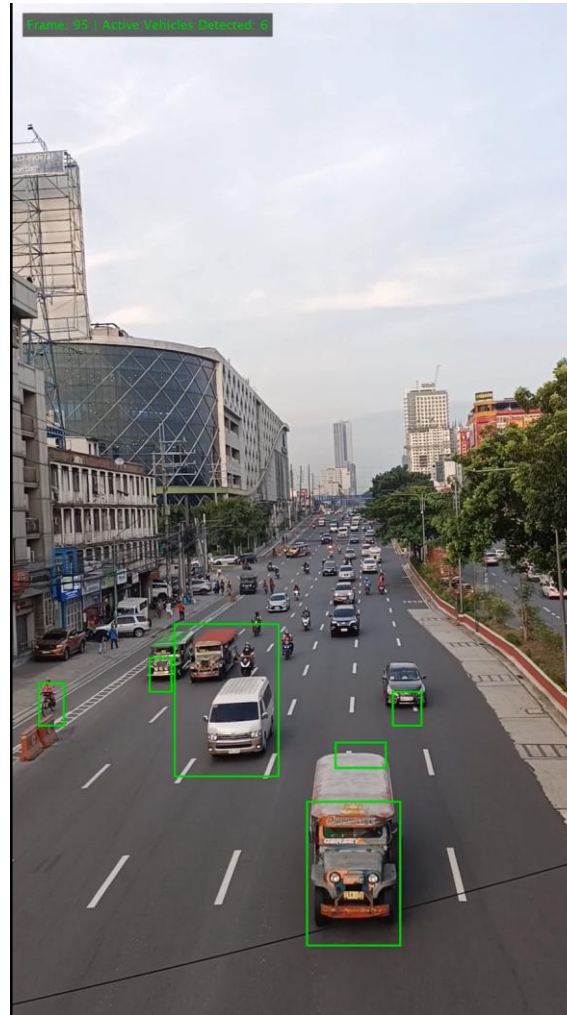
- Background Subtraction
- Line Buffer
- Sobel Edge Detection
- Bounding Box Generator
- Top-Level Tracking

# Results

Input Frame



Output Frame



Simulation Outputs :

- Foreground detection
- Bounding box coordinates
- Processed pixel output

# Technical Skills Demonstrated

- RTL Design
- Verilog HDL Development
- FPGA-Oriented Image Processing
- Functional Verification
- MATLAB/Simulink Integration
- Modular Hardware Design

## Future Scope

- Hardware deployment on FPGA
- Performance optimization
- Resource utilization analysis